

Module 12 - chapter 24

Complete CI/CD with Terraform - Part 3

Final modifications and execute pipeline

We have this code in the 'deploy' stage

```
67
68     sshagent(['server-ssh-key']) {
69         sh "scp -o StrictHostKeyChecking=no server-cmds.sh ${ec2Instance}:/home/ec2-user"
70         sh "scp -o StrictHostKeyChecking=no docker-compose.yaml ${ec2Instance}:/home/ec2-user"
71         sh "ssh -o StrictHostKeyChecking=no ${ec2Instance} ${shellCmd}"
72     }
73 }
```

And we first copy the shell file and then we copy the docker-compose file into EC2 instance and then we ssh into the EC2.

Q:

I am confused how the copy is done if we are not ssh into EC2 at that point?

A:

scp (Secure Copy Protocol) doesn't require you to be interactively logged in first. It's a standalone command that runs from your Jenkins agent and uses SSH under the hood to push files to the remote machine — no prior login needed.

The pattern on line 69:

scp ... server-cmds.sh \${ec2Instance}:/home/ec2-user

means: "from here, copy this file to that remote path" — all in one shot.

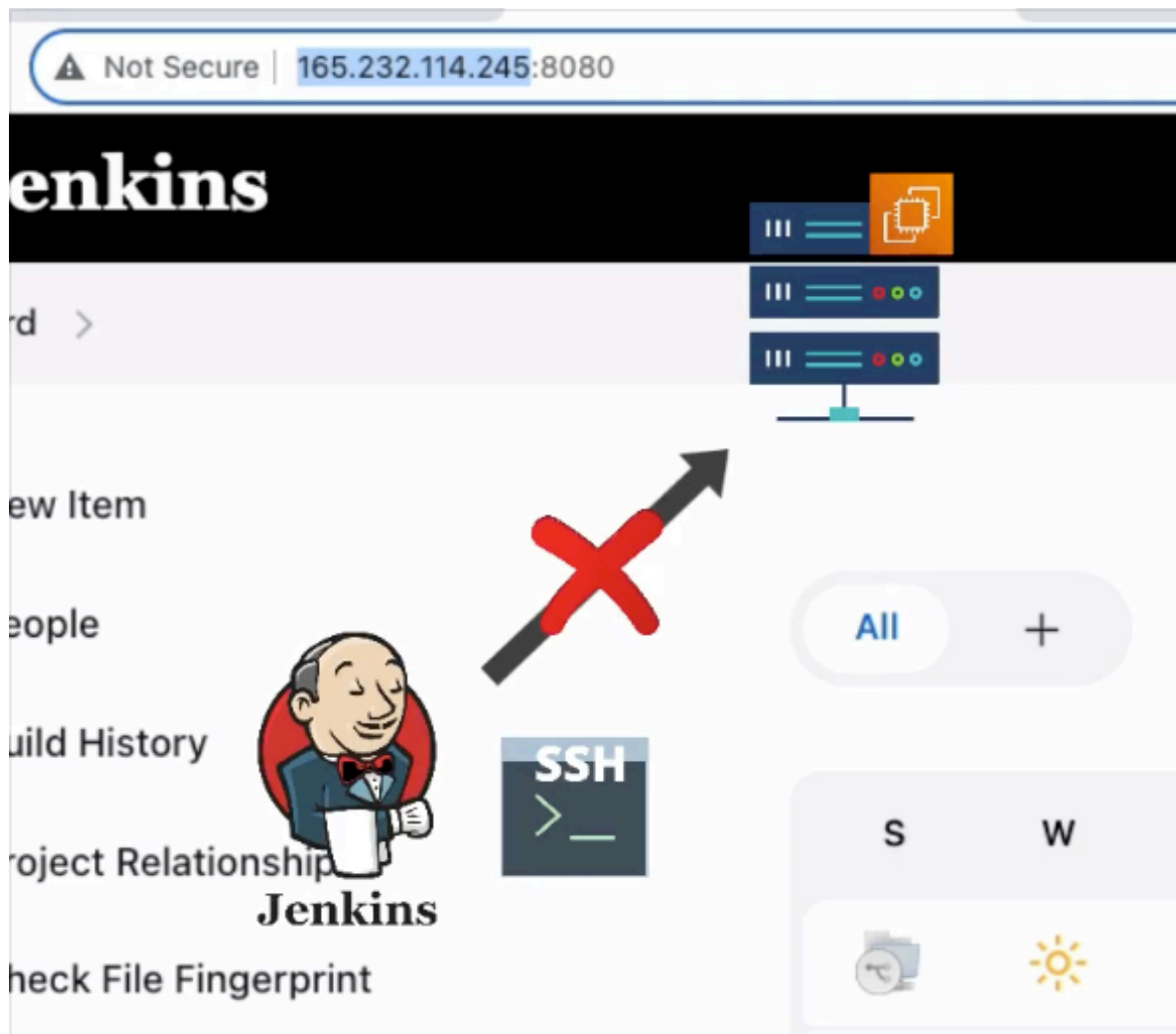
So the sequence is:

1. Lines 69-70 — Jenkins agent uses scp to push files to EC2 (SSH handshake happens internally, file is transferred, connection closes)
2. Line 71 — Jenkins agent opens an interactive SSH session to EC2 and runs \${shellCmd} (likely something like bash /home/ec2-user/server-cmds.sh)

The sshagent(['server-ssh-key']) block on line 68 is what makes all three commands work — it injects the SSH private key into the environment so both scp and ssh can authenticate without a password prompt.

Think of scp like curl — it opens a connection, does its job, and closes. You never "enter" the remote machine; you just reach into it.

Now, Jenkins public IP address doesn't have permissions to ssh into the EC2 instance



We have set access for our own IP address -> my_ip and we need to add the Jenkins IP:

```
terraform > terraform > variables.tf
Jenkinsfile x variables.tf x docker-compose.yaml x
1  +variable vpc_cidr_block {default = "10.0.0.0/16"}
4  +variable subnet_cidr_block {default = "10.0.10.0/24"}
7  +variable avail_zone {default = "eu-west-2a"}
10 +variable env_prefix {default = "dev"}
13 -variable my_ip {
14     default = "140.228.47.252"
15 }
16 -variable jenkins_ip {
17     default = "" You, Moments ago • Uncommitted changes
18 }
19 +variable instance_type {default = "t2.micro"}
22 +variable region {default = "eu-west-2"}
```

And I need to make sure I fixate the IP because it is a cidr block, so I use /32 to fixate it

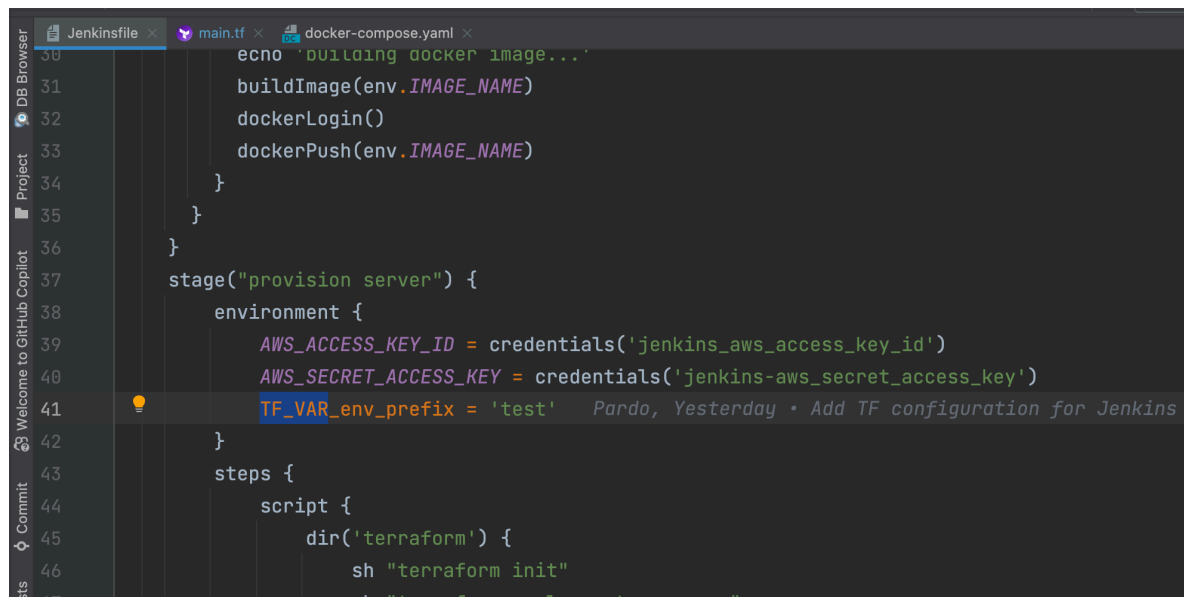
```
13 -variable my_ip {
14     default = "140.228.47.252/32"
15 }
16 -variable jenkins_ip {
17     default = "165.22.116.21/32" You, Moments ago • Uncomm
18 }
19 +variable instance_type {default = "t2.micro"}
```

And now we have to add Jenkins IP address to the security group configuration

```
terraform > terraform > main.tf
Jenkinsfile x main.tf x docker-compose.yaml x
37 }
38 }
39 }
40 resource "aws_default_security_group" "default-sg" {
41     # name = "myapp-sg"
42     vpc_id = aws_vpc.myapp-vpc.id
43 }
44 ingress {
45     from_port = 22
46     to_port = 22
47     protocol = "TCP"
48     cidr_blocks = [var.my_ip, var.jenkins_ip] You, Yesterday • Uncommitted changes
49 }
50 }
jenkins_ip
Press ^, to choose the selected (or first) suggestion and insert a dot afterwards Next Tip
```

So now Jenkins IP address can ssh into EC2 via port 22 🤖

If the Jenkins IP was dynamic then we can use a default value and then add a `TF_VAR_<name_variable>` in `jenkinsfile` like we did for to override the env 'dev' to 'test':

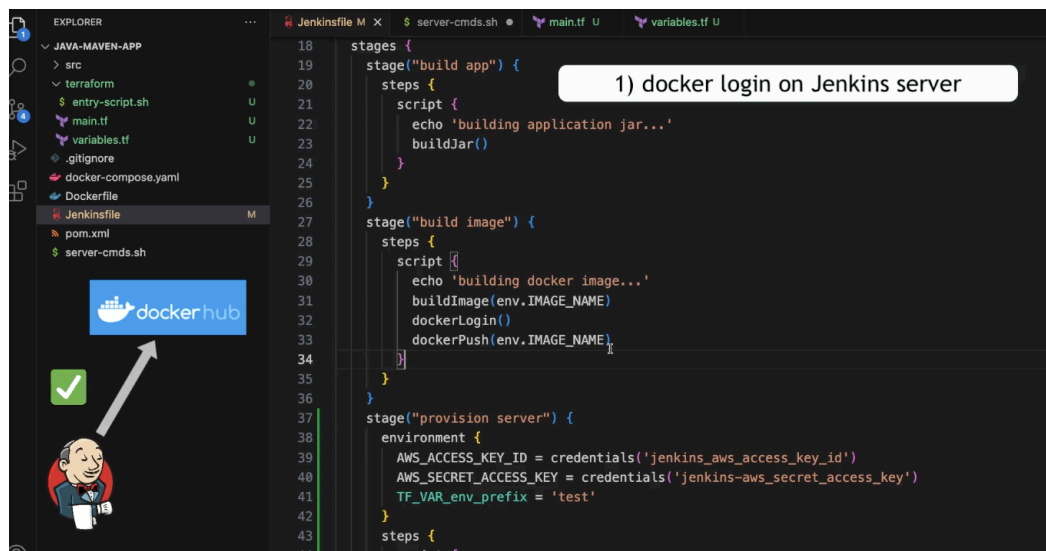


```
30      echo 'building docker image...'
31      buildImage(env.IMAGE_NAME)
32      dockerLogin()
33      dockerPush(env.IMAGE_NAME)
34    }
35  }
36 }
37 stage("provision server") {
38   environment {
39     AWS_ACCESS_KEY_ID = credentials('jenkins_aws_access_key_id')
40     AWS_SECRET_ACCESS_KEY = credentials('jenkins-aws_secret_access_key')
41     TF_VAR_env_prefix = 'test'
42   }
43   steps {
44     script {
45       dir('terraform') {
46         sh "terraform init"
```

Docker login to pull docker image

We need to execute docker login to be able to **pull** images from our repo.

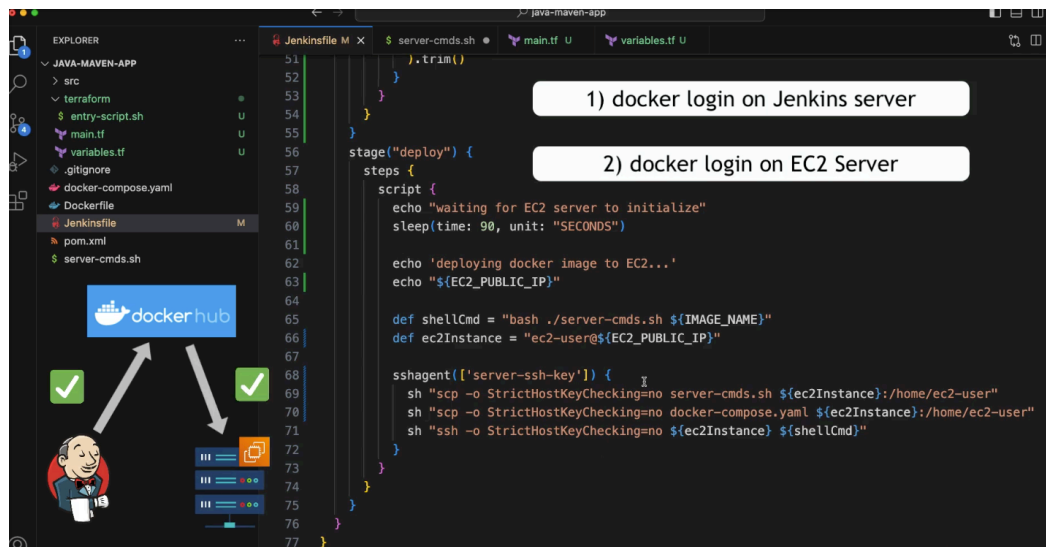
To push the images to the docker repository we also execute docker login in the 'build image' stage but this part doesn't involve EC2 because we docker login on Jenkins Server:



```
18 stages {
19   stage("build app") {
20     steps {
21       script {
22         echo 'building application jar...'
23         buildJar()
24       }
25     }
26   }
27   stage("build image") {
28     steps {
29       script {
30         echo 'building docker image...'
31         buildImage(env.IMAGE_NAME)
32         dockerLogin()
33         dockerPush(env.IMAGE_NAME)
34       }
35     }
36   }
37   stage("provision server") {
38     environment {
39       AWS_ACCESS_KEY_ID = credentials('jenkins_aws_access_key_id')
40       AWS_SECRET_ACCESS_KEY = credentials('jenkins-aws_secret_access_key')
41       TF_VAR_env_prefix = 'test'
42     }
43     steps {
44       script {
```

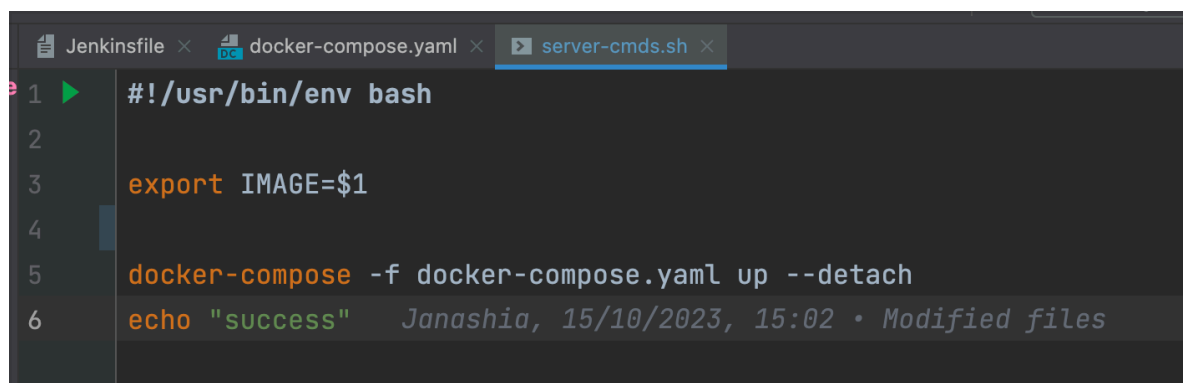
1) docker login on Jenkins server

We need to execute docker login from the EC2 instance so when the server is **pulling** the image from the repository has permission to do it, it can authenticate.



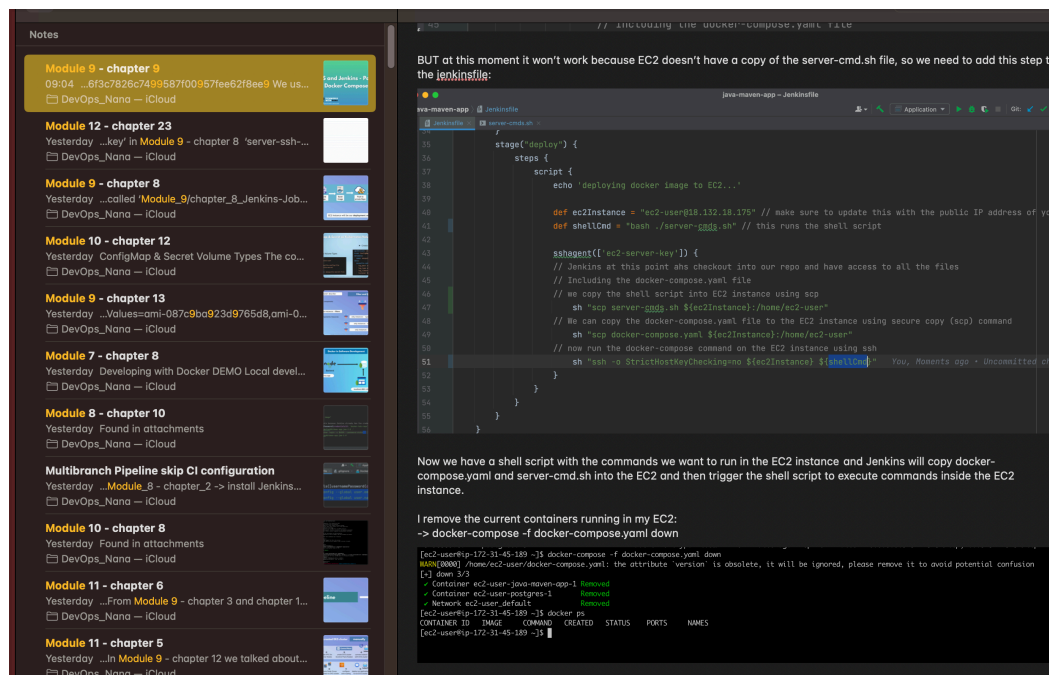
This is the next step -> execute docker login on the EC2 server using jenkinsfile.

For that we will use 'server-cmd.sh' file



Reminder

how the code above was created in Module 9 - chapter 9



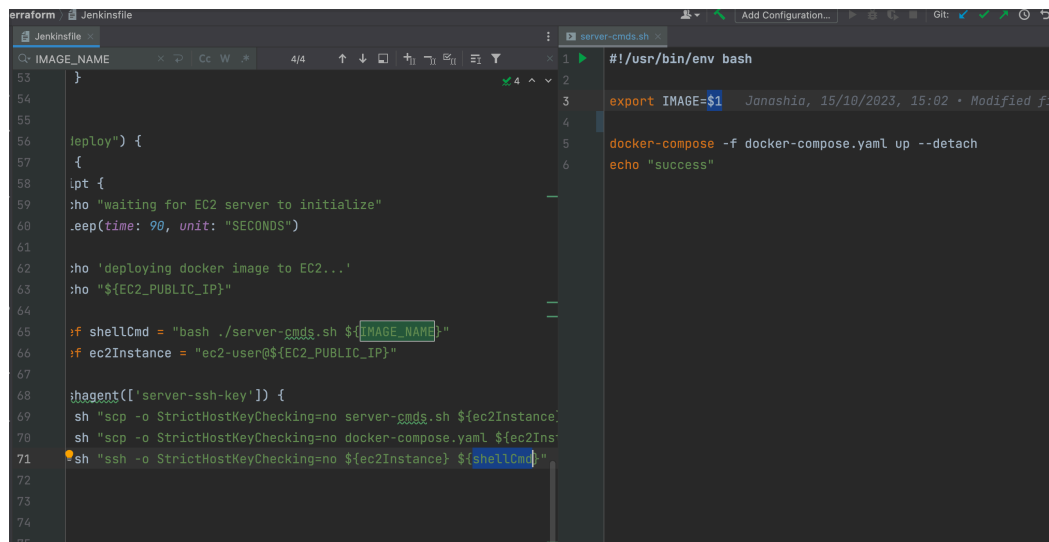
So when we ssh into the EC2 with jenkins file line 71 we also execute shellCmd.

And shellCmd executes "bash ./server-cmds.sh \${IMAGE_NAME}"

Which means run ./server-cmds.sh file and pass parameter IMAGE_NAME to the file so the file can use it.

Because there is only 1 parameter passed in shellCmd then that parameter is \$1.

And in server-cmds.sh the first line (line 3) set parameter \$1 to an env var -> so once this line is executed the EC2 server will have the IMAGE_NAME available globally to be used.



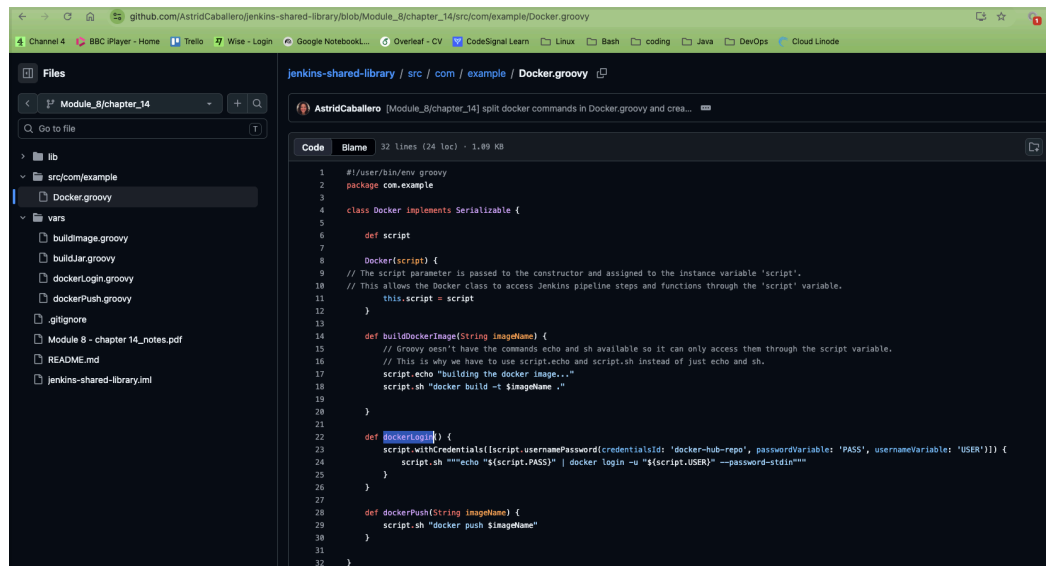
```
jenkinsfile
server-cmds.sh
1 #!/usr/bin/env bash
2
3 export IMAGE=${IMAGE_NAME}
4
5 docker-compose -f docker-compose.yaml up --detach
6 echo "success"
7
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53 }
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56 deploy() {
57 {
58 ipt {
59 :ho "waiting for EC2 server to initialize"
60 .leep(time: 90, unit: "SECONDS")
61
62 :ho 'deploying docker image to EC2...'
63 :ho "${EC2_PUBLIC_IP}"
64
65 :if shellCmd = "bash ./server-cmds.sh ${IMAGE_NAME}"
66 :if ec2Instance = "ec2-user@${EC2_PUBLIC_IP}"
67
68 :hagent(['server-ssh-key']) {
69 sh "scp -o StrictHostKeyChecking=no server-cmds.sh ${ec2Instance}"
70 sh "scp -o StrictHostKeyChecking=no docker-compose.yaml ${ec2Instance}"
71 sh "ssh -o StrictHostKeyChecking=no ${ec2Instance} ${shellCmd}"
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```

```
Jenkinsfile x server-cmds.sh
54
55
56 } } {
57
58 t {
59   "waiting for EC2 server to initialize"
60   p(time: 90, unit: "SECONDS")
61
62   "deploying docker image to EC2..."
63   "${EC2_PUBLIC_IP}"
64
65   shellCmd = "bash ./server-cmds.sh ${IMAGE_NAME}"
66   ec2Instance = "ec2-user@${EC2_PUBLIC_IP}"
67
68   agent(['server-ssh-key']) {
69     "scp -o StrictHostKeyChecking=no server-cmds.sh ${ec2Instance}:"
70     "scp -o StrictHostKeyChecking=no docker-compose.yaml ${ec2Instance}:"
71     "ssh -o StrictHostKeyChecking=no ${ec2Instance} ${shellCmd}"
72
73
```

When I docker login from jenkins to push the image

```
terraform > Jenkinsfile
Jenkinsfile x
21
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34
script {
  echo 'building application jar...'
  buildJar()
}
}
stage("build image") {
  steps {
    script {
      echo 'building docker image...'
      buildImage(env.IMAGE_NAME)
      dockerLogin()
      dockerPush(env.IMAGE_NAME)
    }
  }
}
```

Jenkinsfile made a call to the Jenkins-shared-library -> dockerLogi.groovy -> Docker.groovy -> dockerLogin()



We can see that dockerLogin method needs a password and the user, so we need the same for EC2 but via the server-cmds.sh file

-> echo \$DOCKER_PWD | docker login -u \$DOCKER_USER --password-stdin

This pipes a password into docker login without exposing it in the terminal.
Breaking it down:

echo \$DOCKER_PWD

Prints the Docker password (stored as an environment variable/Jenkins credential) to stdout.

|

Pipes that output as input to the next command.

docker login -u \$DOCKER_USER --password-stdin

- -u \$DOCKER_USER — your Docker Hub username

- --password-stdin — tells Docker to read the password from stdin (the pipe) instead of as a flag argument

Why not just docker login -u \$DOCKER_USER -p \$DOCKER_PWD?

Because passing passwords as flags is a security risk — they show up in:

- Shell history
- Process list (ps aux)
- CI/CD logs

Using --password-stdin keeps the password out of all of those. The pipe feeds it directly into the process's stdin, which never gets logged.

So the full flow is:

\$DOCKER_PWD → echo → pipe → docker reads it as password → authenticated

```

server-cmds.sh x
1  #!/usr/bin/env bash
2
3  export IMAGE=$1
4
5  echo $DOCKER_PWD | docker login -u $DOCKER_USER --password-stdin
6  docker-compose -f docker-compose.yaml up --detach
7  echo "success"

```

Now we need to create DOCKER_PWD and DOCKER_USER, set values to them, then pass them as parameters to server-cmds.sh file and turn them into env vars that line 5 can use to docker login.

```

server-cmds.sh x
1  #!/usr/bin/env bash
2
3  export IMAGE=$1
4  export DOCKER_USER=$2
5  export DOCKER_PWD=$3
6  echo $DOCKER_PWD | docker login -u $DOCKER_USER --password-stdin
7  docker-compose -f docker-compose.yaml up --detach
8  echo "success"

```

So we will pass from jenkinsfile shellCmd with 3 parameters in total
 -> the second parameter will be linked to DOCKER_USER
 -> the third parameter will be linked to DOCKER_PWD

How do we get jenkins to pass EC2 the credentials?
 We have the dockerhub credentials in jenkins

Jenkins / Manage Jenkins / Credentials

Credentials

T	P	Store ↓	Domain	ID	Name
		System	(global)	github-ssh-credentials	AstridCaballero
		System	(global)	docker-hub-repo	cabfish00/*****
		System	(global)	nexus-docker-repo	astrid/*****
		System	(global)	server-credentials	server-user/*****

So let's create an environment block in the 'deploy' stage:

credentials('docker-hub-repo') -> Returns the whole object

```

56 stage("deploy") {
57     environment {
58         DOCKER_CREDS = credentials('docker-hub-repo')
59     }
60     steps {
61         script {
62             echo "waiting for EC2 server to initialize"
63             sleep(time: 90, unit: "SECONDS")
64
65             echo 'deploying docker image to EC2...'
66             echo "${EC2_PUBLIC_IP}"
67
68             def shellCmd = "bash ./server-cmds.sh ${IMAGE_NAME}"
69             def ec2Instance = "ec2-user@${EC2_PUBLIC_IP}"
70
71             sshagent(['server-ssh-key']) {
72                 sh "scp -o StrictHostKeyChecking=no server-cmds.sh ${ec2Instance}"
73                 sh "scp -o StrictHostKeyChecking=no docker-compose.yaml ${ec2Instance}"
74                 sh "ssh -o StrictHostKeyChecking=no ${ec2Instance} ${shellCmd}"
75             }
76         }
77     }
78 }

```

When extracting credentials with 'username with password' type

Global credentials (unrestricted)		
Credentials that should be available irrespective of domain specification to requirements matching.		
ID	Name	Kind
 github-ssh-credentials	AstridCaballero	SSH Username with private key
 docker-hub-repo	cabfish00/*****	Username with password
 nexus-docker-repo	astrid/*****	Username with password
 server-credentials	server-user/*****	Username with password

Then we get additional env vars set automatically with user and password prefixes, so we can get the user with DOCKER_CREDS_USR

```

56 stage("deploy") {
57     environment {
58         DOCKER_CREDS = credentials('docker-hub-repo')
59     }
60     steps {
61         script {
62             echo "waiting for EC2 server to initialize"
63             sleep(time: 90, unit: "SECONDS")
64
65             echo 'deploying docker image to EC2...'
66             echo "${EC2_PUBLIC_IP}"
67
68             def shellCmd = "bash ./server-cmds.sh ${IMAGE_NAME} ${DOCKER_CREDS_USR}"
69             def ec2Instance = "ec2-user@${EC2_PUBLIC_IP}"
70
71             sshagent(['server-ssh-key']) {
72                 sh "scp -o StrictHostKeyChecking=no server-cmds.sh ${ec2Instance}"
73                 sh "scp -o StrictHostKeyChecking=no docker-compose.yaml ${ec2Instance}"
74                 sh "ssh -o StrictHostKeyChecking=no ${ec2Instance} ${shellCmd}"
75             }
76         }
77     }
78 }

```

So we pass the second parameter (\$2) to the shellCmd as seen in line 68 🙌

And the prefix for the password is PSW -> DOCKER_CREDS_PSW

Variable	Contains
DOCKER_CREDS	username:password (colon-separated)
DOCKER_CREDS_USR	username only
DOCKER_CREDS_PSW	password only

The rule is simple: whatever you name your variable + _USR / _PSW. If you had named it MY_SECRET, you'd get MY_SECRET_USR and MY_SECRET_PSW.

Where to find the docs:

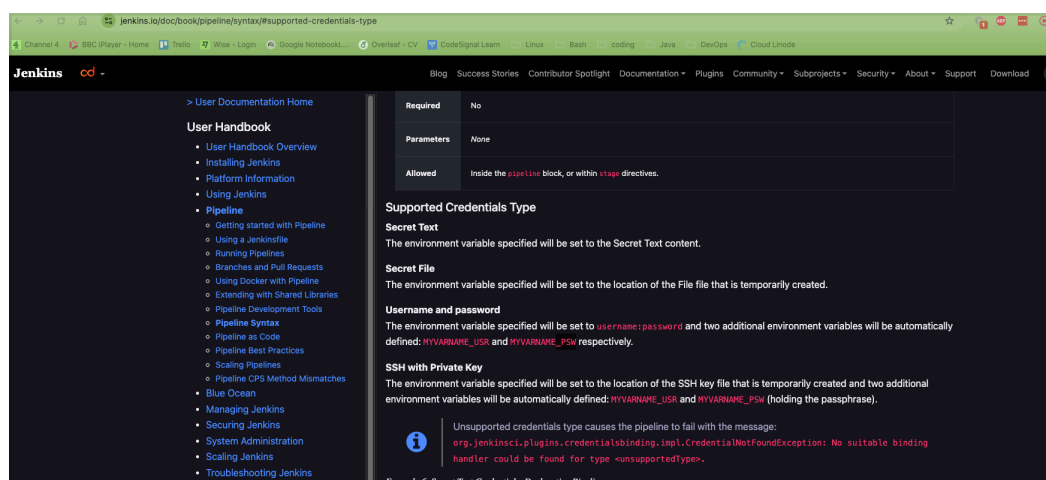
Search for "Jenkins Credentials Binding" or go to jenkins.io and look for:

- Pipeline > Using credentials in the official Jenkins docs
- The Credentials Binding Plugin page on the Jenkins plugin site (plugins.jenkins.io)

The specific section to look for is "Username and password" under credential types — it explicitly lists the _USR and _PSW suffixes that get auto-generated.

This only works for "Username with password" credential type in Jenkins. Other credential types (secret text, SSH key, etc.) give you different variables.

<https://www.jenkins.io/doc/book/pipeline/syntax/#supported-credentials-type>



So now we are passing 3 parameters in shellCmd


```

56 stage("deploy") {
57     environment {
58         DOCKER_CREDS = credentials('docker-hub-repo')
59     }
60     steps {
61         script {
62             echo "waiting for EC2 server to initialize"
63             sleep(time: 90, unit: "SECONDS")
64
65             echo 'deploying docker image to EC2...'
66             echo "${EC2_PUBLIC_IP}"
67
68             def shellCmd = "bash ./server-cmds.sh ${IMAGE_NAME} ${DOCKER_CREDS_USR} ${DOCKER_CREDS_PSW}"
69             def ec2Instance = "ec2-user@${EC2_PUBLIC_IP}"
70
71             sshagent(['server-ssh-key']) {
72                 sh "scp -o StrictHostKeyChecking=no server-cmds.sh ${ec2Instance}:/home/ec2-user"
73                 sh "scp -o StrictHostKeyChecking=no docker-compose.yaml ${ec2Instance}:/home/ec2-user"
74                 sh "ssh -o StrictHostKeyChecking=no ${ec2Instance} ${shellCmd}"
75             }

```

Now when jenkinsfile executes the ssh command in line 74 to ssh into the EC2 instance and then executes shellCmd variable it will execute the commands in the server-cmds.sh file and EC2 will be able to docker login hence docker-compose.yaml will be able to pull our docker image from docker hub and work as expected.

```

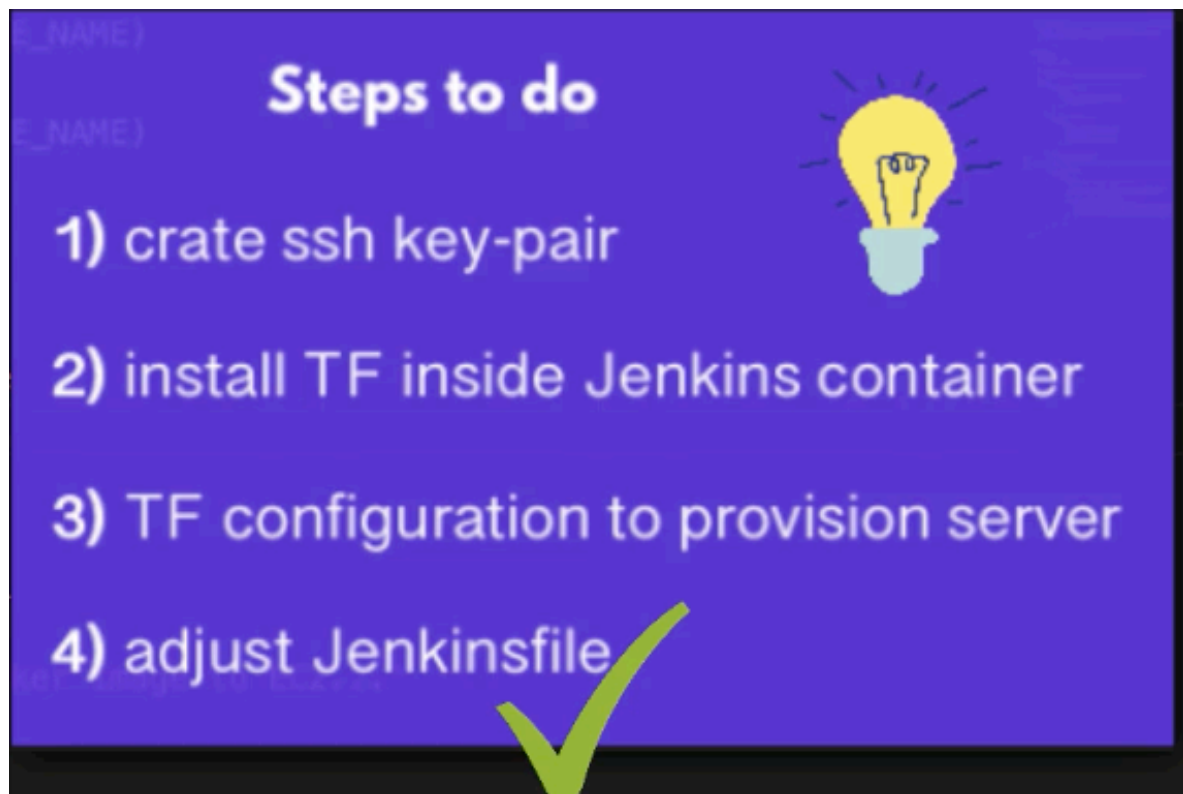
Jenkinsfile
54 stage("deploy") {
55     environment {
56         DOCKER_CREDS = credentials('docker-hub-repo')
57     }
58     steps {
59         script {
60             echo "waiting for EC2 server to initialize"
61             sleep(time: 90, unit: "SECONDS")
62
63             echo 'deploying docker image to EC2...'
64             echo "${EC2_PUBLIC_IP}"
65
66             def shellCmd = "bash ./server-cmds.sh ${IMAGE_NAME} ${DOCKER_CREDS_USR} ${DOCKER_CREDS_PSW}"
67             def ec2Instance = "ec2-user@${EC2_PUBLIC_IP}"
68
69             sshagent(['server-ssh-key']) {
70                 sh "scp -o StrictHostKeyChecking=no server-cmds.sh ${ec2Instance}:/home/ec2-user"
71                 sh "scp -o StrictHostKeyChecking=no docker-compose.yaml ${ec2Instance}:/home/ec2-user"
72                 sh "ssh -o StrictHostKeyChecking=no ${ec2Instance} ${shellCmd}"
73             }
74         }
75     }
76 }

```

```

server-cmds.sh
1 #!/usr/bin/env bash
2
3 export IMAGE=$1
4 export DOCKER_USER=$2
5 export DOCKER_PWD=$3
6 echo $DOCKER_PWD | docker login -u $DOCKER_USER --password-stdin
7 docker-compose -f docker-compose.yaml up --detach
8 echo "success"

```



Run CI/Cd pipeline

I push the changes and run the pipeline but it failed

```
Jenkins / terraform-pipeline / Module_12/Terraform_chapter... / #2 / Console Output

[32m+ [0m [0m tags_all = {
[32m+ [0m [0m "Name" = "test-vpc"
}

[1mPlan: [0m [0m5 to add, 0 to change, 0 to destroy.
[31m, [0m [0m
[31m] [0m [0m [1m [31mError: [0m [0m [1mreading SSM Parameter [/aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64]: operation error SSM: GetParameter, https
response error StatusCode: 400, RequestID: 946efe43-d1aa-4897-9dbc-9a83a8f0642b, api error AccessDeniedException: User: arn:aws:iam::706976333940:user/jenkins is not
authorized to perform: ssm:GetParameter on resource: arn:aws:ssm:eu-west-2::parameter/aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 because no identity-
based policy allows the ssm:GetParameter action [0m
[31m] [0m [0m
[31m] [0m [0m [0m with data.aws_ssm_parameter.latest-amazon-linux-image,
[31m] [0m [0m [0m on main.tf line 72, in data "aws_ssm_parameter" "latest-amazon-linux-image":
[31m] [0m [0m 72: data "aws_ssm_parameter" "latest-amazon-linux-image" [4m( [0m [0m
[31m] [0m [0m
[31m] [0m [0m
```

AI explains it:

Clear permissions error. Jenkins is running Terraform as IAM user jenkins, but that user lacks the `ssm:GetParameter` permission needed to look up the latest Amazon Linux AMI.

What's happening in your Terraform:

hcl

```
# main.tf line 72 — fetching latest AL2023 AMI from AWS's public SSM path
data "aws_ssm_parameter" "latest-amazon-linux-image" {
  name = "/aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-
x86_64"
}
```

AWS stores public AMI IDs in SSM Parameter Store, and Terraform queries it — but your Jenkins IAM user can't call `ssm:GetParameter`.

Fix — attach this policy to the jenkins IAM user in AWS:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["ssm:GetParameter"],
      "Resource": ["arn:aws:ssm:eu-west-2::parameter/aws/service/*"]
    }
  ]
}
```

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["ssm:GetParameter"],
      "Resource": ["arn:aws:ssm:eu-west-2::parameter/aws/service/*"]
    }
  ]
}
```

How to apply it:

1. AWS Console → IAM → Users → jenkins
2. Permissions → Add permissions → Attach policies directly → Create inline policy
3. Paste the JSON above

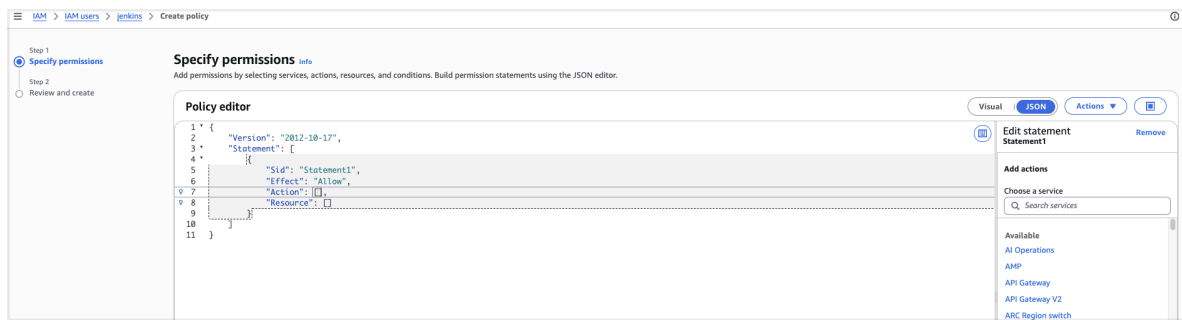
Or via AWS CLI:

```
aws iam put-user-policy \
  --user-name jenkins \
  --policy-name ssm-read-ami \
  --policy-document file://policy.json
```

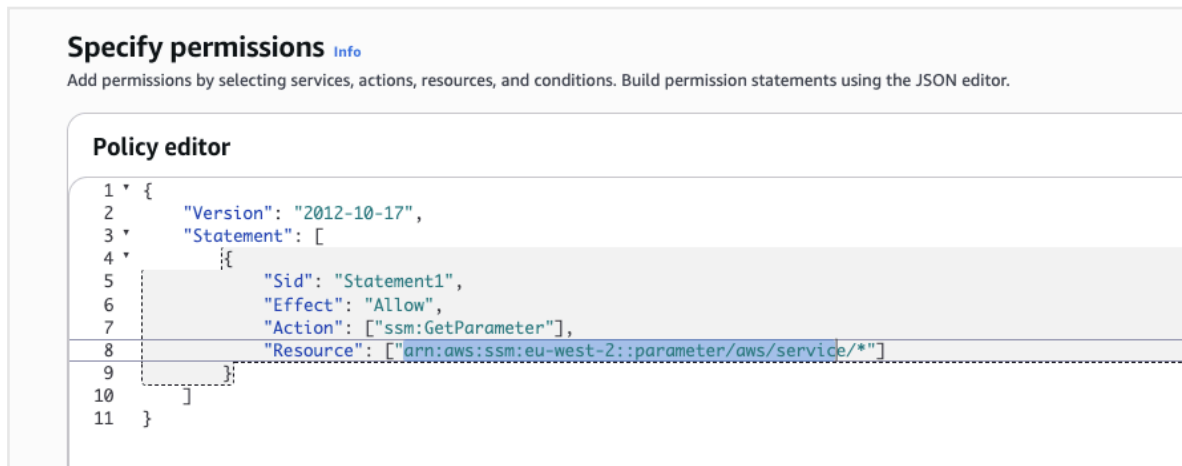
The resource `arn:aws:ssm:eu-west-2::parameter/aws/service/*` scopes it only to AWS's public parameter paths (AMIs, etc.) — not your own SSM secrets.

So this happened because I used SSM in my code, Nana didn't use it. And as the AI explained jenkins is using its own user called 'jenkins' and I need to edit this user to be able to use SSM. Nana used 'admin' user who has more permissions.

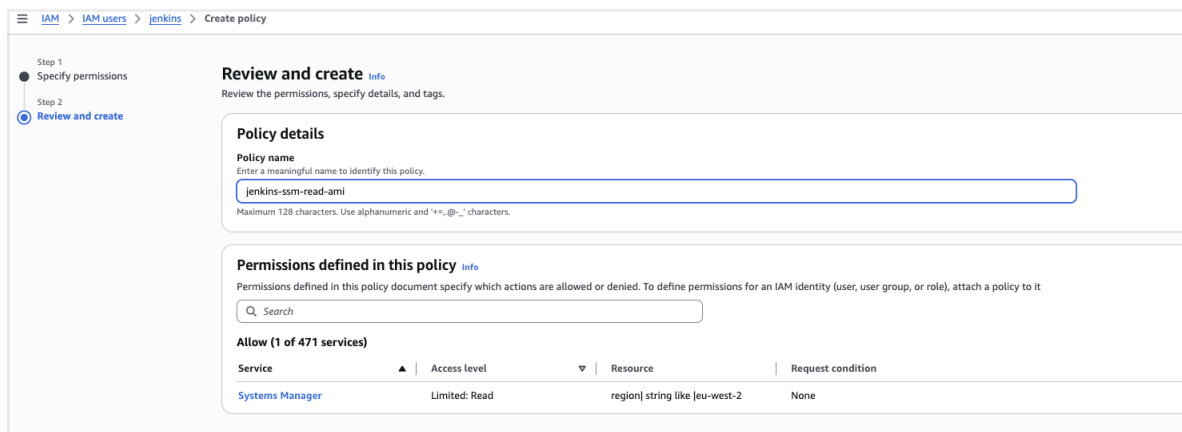
And I will fix it via the AWS console



Now I edit the action and resource values in lines 7 and 8:



Then next and I give the policy a name 'jenkins-ssm-read-ami'



And jenkins now has the SSM permission

Q:

and with that policy then I don't need the ssm policy?

A:

Actually you do still need the SSM policy — `ec2:*` only covers EC2 actions, it doesn't include SSM.

AmazonEC2FullAccess (the AWS managed policy) actually does bundle SSM read access in it though, so if you went that route you could drop the custom SSM policy.

But if you wrote the `ec2:*` JSON manually, keep both policies attached.

So I am going to delete the SSM policy I just created and use 'AmazonEC2FullAccess' instead.

1. AWS Console → IAM → Users → click on jenkins
2. Permissions tab → Add permissions button
3. Select "Attach policies directly"
4. In the search box type AmazonEC2FullAccess
5. Check the checkbox next to it
6. Click Next → Add permissions

Add permissions
Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

- ☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
- ☐ Copy permissions
Copy all group memberships, attached managed policies, inline policies, and any existing permissions boundaries from an existing user.
- ☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1509)

Filter by Type: All types 1 match

Policy name	Type	Attached entities
☒ AmazonEC2FullAccess	AWS managed	1

Cancel Next

Review
The following policies will be attached to this user. [Learn more](#)

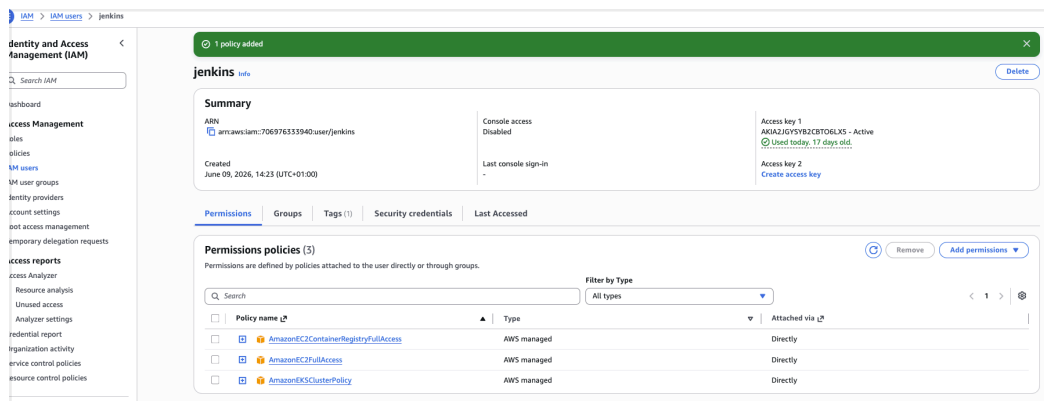
User details

User name
jenkins

Permissions summary (1)

Name	Type	Used as
AmazonEC2FullAccess	AWS managed	Permissions policy

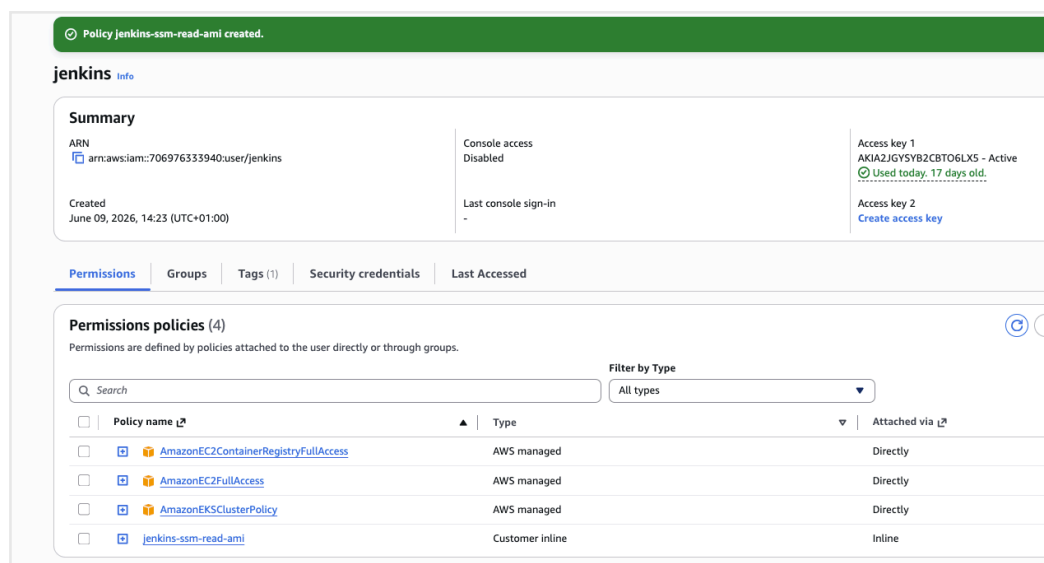
Cancel Previous **Add permissions**



Re-build again and it fails
Because of SSM



So I will the policy back



Re-build pipeline -> it passed 'provision server' stage but it fail 'deploy' stage

```
[Pipeline] echo
waiting for EC2 server to initialize
[Pipeline] sleep
Sleeping for 1 min 30 sec
[Pipeline] echo
deploying docker image to EC2...
[Pipeline] echo
"18.171.52.103"
[Pipeline] sshagent
[ssh-agent] Using credentials ec2-user
$ ssh-agent
SSH_AUTH_SOCK=/tmp/ssh-6yTbzPcjv71z/agent.104399
SSH_AGENT_PID=104404
Running ssh-add (command line suppressed)
Error loading key "/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24/tmp/private_key_15504326899571478610.key": error in libcrypto
[Pipeline] // sshagent
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // withEnv
```

AI explains:

This is a key format issue. Jenkins is failing to load the private key because it's in OpenSSH format but the version of libcrypto on the Jenkins container expects PEM format.

How to tell the difference:

OpenSSH format (causes this error):

-----BEGIN OPENSSH PRIVATE KEY-----

PEM format (what Jenkins needs):

-----BEGIN RSA PRIVATE KEY-----

Fix — convert your key to PEM format on your local machine:

```
ssh-keygen -p -m PEM -f ~/.ssh/your-key-file
```

This converts it in place. It will ask for a passphrase — just press Enter if your key has none.

Then:

1. Copy the contents of the converted private key
2. AWS Console → IAM → back to Jenkins credentials in Jenkins (Manage Jenkins → Credentials)
3. Find your server-ssh-key credential → update it with the new PEM formatted key
4. Re-run the pipeline

Alternatively, generate a fresh key pair in PEM format from the start:

```
ssh-keygen -t rsa -b 4096 -m PEM -f ~/.ssh/new-jenkins-key
```

Then update both the Jenkins credential and the public key on your EC2 instance.

So I checked but the format is correct


```

> cd ..
> cd Downloads
> cat myapp-key-pair.pem
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAz2awq7nFnGzG8ZrRnWFKqCR8yyAfE0
tuiPPpY06SW1c29Y9JyqttXRSKBgwNym9g0MJsvQ6VIq6L

```

Then I looked at the screenshot at the moment I created the credential in jenkins and I found the problem -> a TYPO 🥲

I included the % at the end -> Module 12 - chapter 23

Jenkins / terraform-pipeline / Credentials / Folder / Global credentials (unrestrict...)

New credentials

Kind
SSH Username with private key

ID ?
server-ssh-key

Description ?

Username
ec2-user

☐ Treat username as secret ?

Private Key
☒ Enter directly

Key

```

v3dLvReE6v0Zwth010XjB8jwRwM0xGjMCuotUi+1UR/Z62TVu5T8jIY4/LYD1B
F2G/ruTd/ltBC1ppeSWF7/MYswpsz8ZMFQe0c9pRAoGBAKbHlyGfLy1oFaJYNG8
nrSMebkNQIjkdUyVYfNF8zUmV25nmNjKKD0EyziTuWCbD0A/HnXX9LyXXhRYrDY
U1MuLZrNSQwmfApB7agEKDSaVFRFr9tLVZiN0Uix8UrgKHyCuyk+tgq+aEcMecLc
jBaHLZmTcjvRrfylnJTPDJBtAoGAZ4+MKJH6150BKvJHeZULiq7UcyX0s/ueWY7B
LCXkKkWAJWyo2yJ+041RZE+0VnCKKhcailzyA3Aj31dhjFtDv0vw+LFE/hpcG/cWf
w02N8YbMVhb1L7EwMDLAPwRL5uSr+1L1agqpS5zkzWq/IOJ1wCDM/zZ5K+rg0y3Z
/gWkuREcGYAx0uGxqansQW4iZTX2ER1UTVCEa+BJ53bpKL7a40AAWq8rc/pztW
h/LwuM7CbbVh0ToxZjLLFDAdaamaKpn134B5JF0qy29rmnIZfjvLDP8W6LF0JxzE
erRLYZs0LTZSYIX75mhEpv00NuWoeby9NR4kgmxLlvY08gr7aU4bTw==
-----END RSA PRIVATE KEY-----

```

Passphrase

Create

So I removed the typo:

Jenkins terraform-pipeline Credentials Folder Global credentials (unrestrict... ec2-user

Update Delete Move

Update credentials

ID ?
server-ssh-key

Description ?

Username
ec2-user

☐ Treat username as secret ?

Private Key
☒ Enter directly

Key

Enter New Secret Below

y3dLvrE6v0Zrthd10XjB8jvfrwM0nGjHCuotUj+1UR/262TVuST8jIY4/LY01B
F2G/rutd/LtBClpeSWF7/RYsmps82FQe0c9pRAoGBAKhlyGfQlv1ofaJYNG8
nr5heblANQjIjK0dyvYfNF8zUmV25m4jKKD0EyzL1uWC3d0A/HnXX9lyXzhRyrdY
UfhuLzNS0wmfAp87anGd5aVFRF rP1LVZJNBULx8Urgkthyclyk+tag+aECfec1c
jBahlZnTCjvRrfynJTPDJBtAoGAZ4+MKJH6158BKvJHeZU1q7UcyX0s/ueky7B
LCXKkWAjwozyJ+041RZE+0vmCKKca1zyA36j31dhjFIDV8vwm1FE/hpG6/CWf
w0ZNB78PmVnIL75mD1APML5u5r+IL1agop5+2kcew/10J1ncDM/z28K+rdy3Z
/gWuREcgyAx0uGxam5Qw4iZTX2ER1UTVCeA+BhU353bpKL7a48AAWqBrc/pztM
h/Lw0H/CbbVh0TosZj\LFDAaamaKpN13485JF8qz59rm1Zj\jvDP8W6LF0JxzE
erR1TZa81TZ51X75mEgV8Wu0eeby8W84gmxLlv188pr7ah4uNm=
-----END RSA PRIVATE KEY-----

Passphrase

Save

And saved it.

Re-build the pipeline and it failed due to error code 127

```

unset SSH_AUTH_SOCK;
unset SSH_AGENT_PID;
echo Agent pid 105443 killed;
[ssh-agent] Stopped.
[Pipeline] // sshagent
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 127

```

The problem is

```

Affected argument(s) used the following variable(s): [DOCKER_CREDS_PSW]
See https://jenkins.io/redirect/groovy-string-interpolation for details.

+ VDC8*p8C
/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24@tmp/durable-b74dcc5b/script.sh.copy: 1: VDC8*p8C: not found
+ ssh -o StrictHostKeyChecking=no ec2-user@18.171.52.103 bash /home/ec2-user/server-cmds.sh cabfish00/demo-app:java-maven-2.0 cabfish00 Mpu
[Pipeline] }
$ ssh-agent -k
unset SSH_AUTH_SOCK;
unset SSH_AGENT_PID;

```

AI explains:

The main problem is this line:

+ VDC8*p8C

VDC8*p8C: not found

Jenkins is treating your Docker password as a shell command. This happens because shellCmd uses a double-quoted Groovy string with \$

{DOCKER_CREDS_PSW} — Groovy interpolates it before passing to the shell.

Also notice the password is truncated in the ssh command:

```
bash /home/ec2-user/server-cmds.sh cabfish00/demo-app:java-maven-2.0  
cabfish00 MpU
```

MpU is not the full password — the * in VDC8*p8C caused the shell to try glob expansion and broke everything.

Fix — change how you build shellCmd:

```
def shellCmd = "bash /home/ec2-user/server-cmds.sh ${IMAGE_NAME} $  
{DOCKER_CREDS_USR} ${DOCKER_CREDS_PSW}"
```

Change to:

```
def shellCmd = 'bash /home/ec2-user/server-cmds.sh ' + IMAGE_NAME + ' ' +  
DOCKER_CREDS_USR + ' ' + DOCKER_CREDS_PSW
```

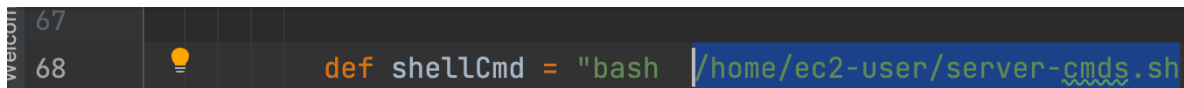
Using single quotes prevents Groovy interpolation, then concatenating the variables passes them safely to the shell.

I also made couple of changes:

Used absolute path in shellCmd to make sure jenkins fetches the pasted server-cmds.sh in EC2

I changed ./server-cmds.sh to /home/ec2-user/server-cmds.sh

./ means current directory and when jenkins ssh into EC2 the current dir might not be /home/ec2-user according to AI.



```
67  
68 def shellCmd = "bash /home/ec2-user/server-cmds.sh
```

At the end I reverted the changes to shellCmd and keep the syntax as Nana's.

AI suggest:

I think I've found the real root cause. Your Docker Hub password VDC8*p8C contains a * character which the shell interprets as a glob wildcard. That's why it's being treated as a separate command instead of an argument.

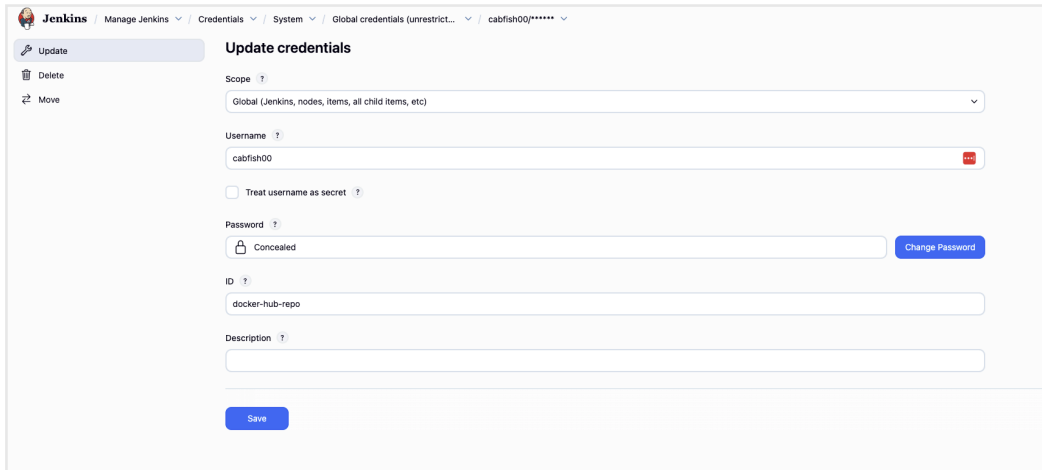
This is a shell special character problem, not a Groovy/Jenkins problem.

The quickest fix is to change your Docker Hub password to one without special shell characters (*, !, \$, &, #, etc.).

1. Go to hub.docker.com → Account Settings → Security → Change Password
2. Set a new password using only letters, numbers, and simple characters like - or _
3. Update the docker-hub-repo credential in Jenkins with the new password

This is actually a common gotcha when passwords are used as shell arguments without quoting — special characters get interpreted by the shell. Once the password has no special characters, your teacher's double-quoted string approach will work fine.

So I reset my dockerhub password and I updated the password in jenkins credentials 'docker-hub-repo' as well:



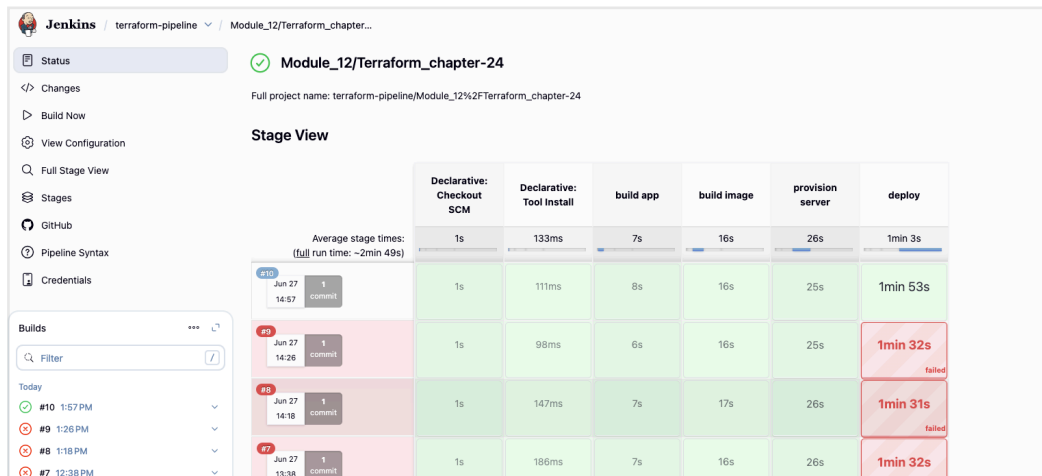
The screenshot shows the Jenkins 'Update credentials' interface. The breadcrumb trail is: Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestrict...) / cabfish00/*****. On the left, there are buttons for 'Update', 'Delete', and 'Move'. The main form has the following fields: 'Scope' (Global (Jenkins, nodes, items, all child items, etc)), 'Username' (cabfish00), 'Treat username as secret' (unchecked), 'Password' (Concealed), 'ID' (docker-hub-repo), and 'Description'. A 'Change Password' button is next to the password field. At the bottom is a 'Save' button.

And now the console shows EC2 logged into docker hub and execute the shellCmd etc

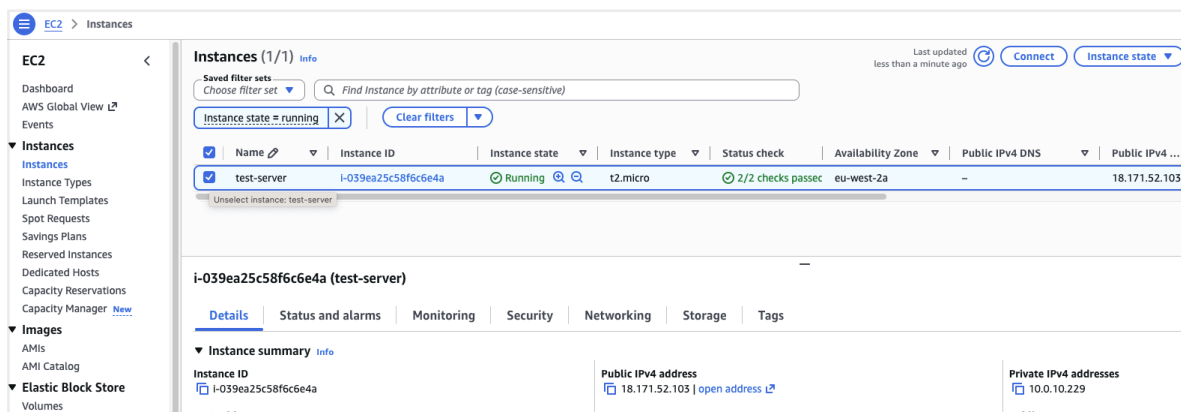
```
[Pipeline] {
[Pipeline] sh
+ scp -o StrictHostKeyChecking=no server-cmds.sh ec2-user@18.171.52.103:/home/ec2-user
[Pipeline] sh
+ scp -o StrictHostKeyChecking=no docker-compose.yaml ec2-user@18.171.52.103:/home/ec2-user
[Pipeline] sh
Warning: A secret was passed to "sh" using Groovy String interpolation, which is insecure.
Affected argument(s) used the following variable(s): [DOCKER_CREDS_PSW]
See https://jenkins.io/redirect/groovy-string-interpolation for details.
+ ssh -o StrictHostKeyChecking=no ec2-user@18.171.52.103 bash ./server-cmds.sh cabfish00/demo-app:java-maven-2.0 cabfish00 ****
WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
time="2026-06-27T14:00:20Z" level=warning msg="/home/ec2-user/docker-compose.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
java-maven-app Pulling
postgres Pulling
9b70e313681f Pulling fs layer
569c90a3d642 Pulling fs layer
```





Now let's check AWS UI



And I see that the EC2 has been initialised.

And we can check EC2 from our laptop:

First let's move the pem file from downloads to .ssh folder

```
> mv ~/Downloads/myapp-key-pair.pem ~/.ssh
> cd ..
> cd .ssh
> ls -l myapp-key-pair.pem
-rw-r--r--@ 1 astrid staff 1678 26 Jun 08:46 myapp-key-pair.pem
```

Then we need to restrict permissions to the pem file locally (like we always do)
-> chmod 400 myapp-key-pair.pem

```
> chmod 400 myapp-key-pair.pem
> ls -l myapp-key-pair.pem
-r-----@ 1 astrid staff 1678 26 Jun 08:46 myapp-key-pair.pem
```

Now let's ssh into the EC2 instance

-> ssh -i ~/.ssh/myapp-key-pair.pem ec2-user@18.171.52.103

```
> ssh -i ~/.ssh/myapp-key-pair.pem ec2-user@18.171.52.103
#_
~\_ #####_ Amazon Linux 2023
~~ \_#####\
~~ \###|
~~ \#/ _____ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~
~~~. _ . _/
~~~/_/_/_/
~~~/_/m/'
[ec2-user@ip-10-0-10-229 ~]$
```

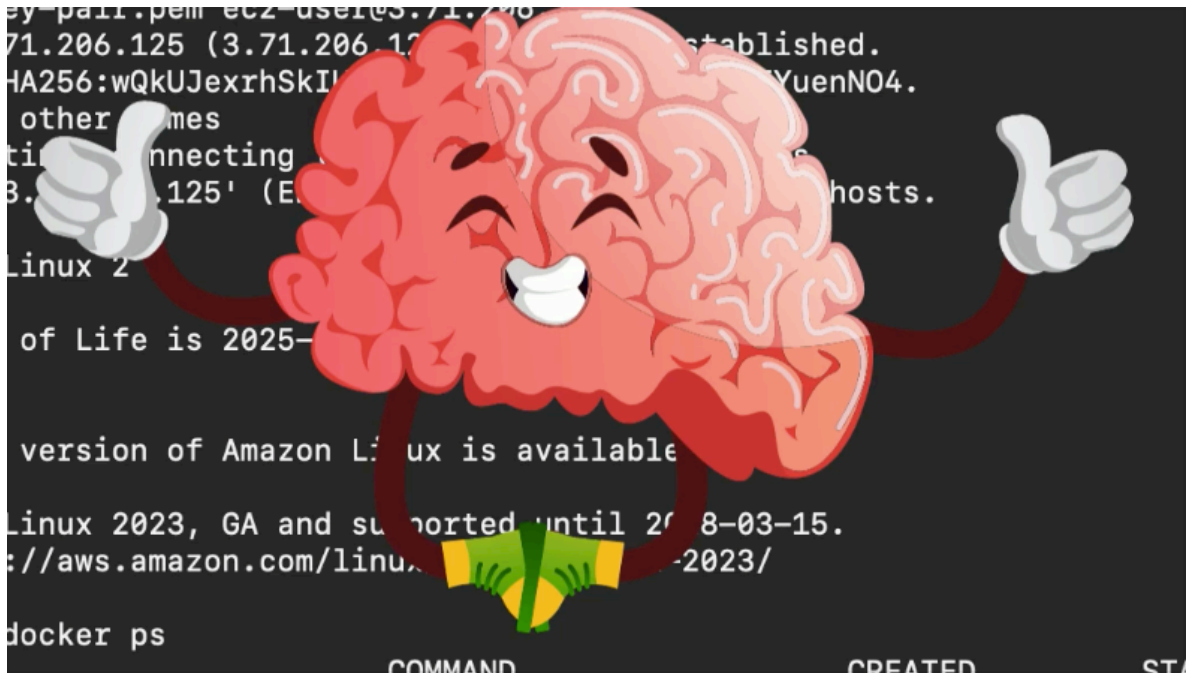
```
-> docker ps
```

```
[ec2-user@ip-10-0-10-229 ~]$ docker ps
```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
b1ca0b12b2ab	cabfish00/demo-app:java-maven-2.0		"java -jar java-mave..."	19 minutes ago	Up 19 minutes	0.0.0.0:8080->8080/t
a051e0fb5a79	ec2-user-java-maven-app-1	postgres:16	"docker-entrypoint.s..."	19 minutes ago	Up 19 minutes	0.0.0.0:5432->5432/t

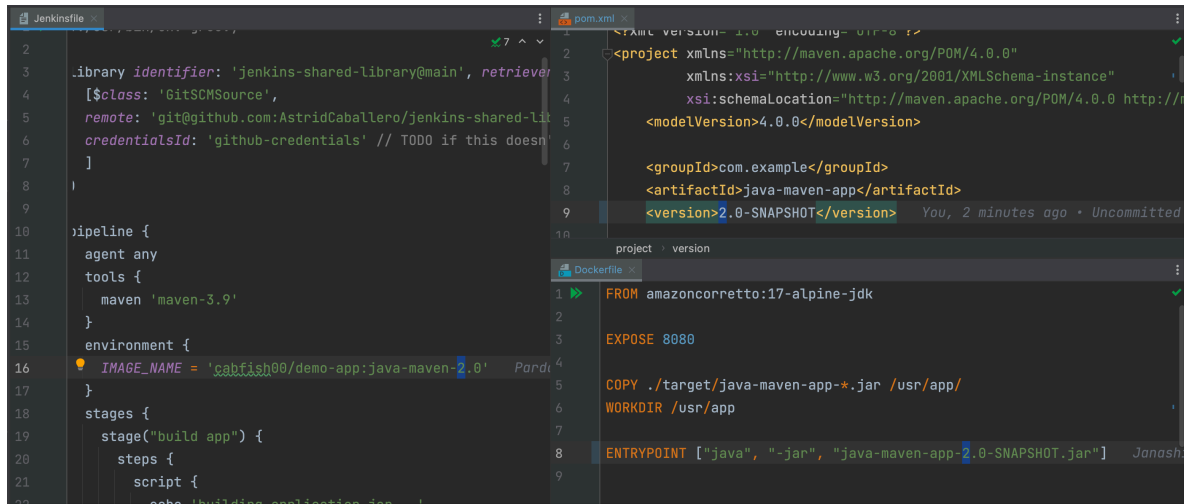
```
p, ::5432->5432/tcp ec2-user-postgres-1
[ec2-user@ip-10-0-10-229 ~]$
```

And we can see that docker-compose run my java app and Postgres, my EC2 instance has 2 containers 🥳

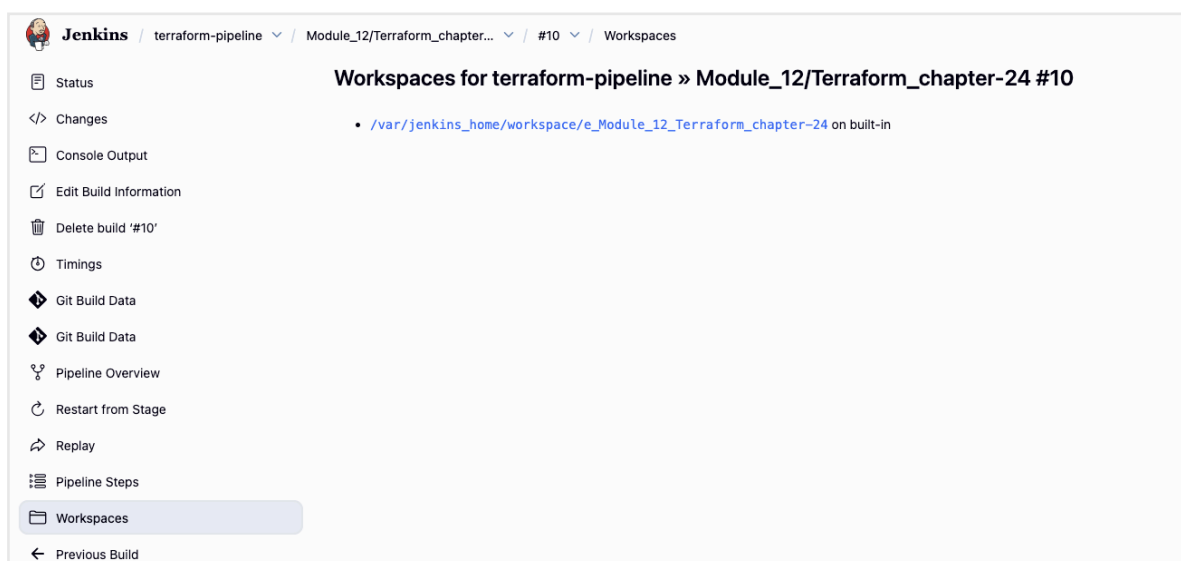
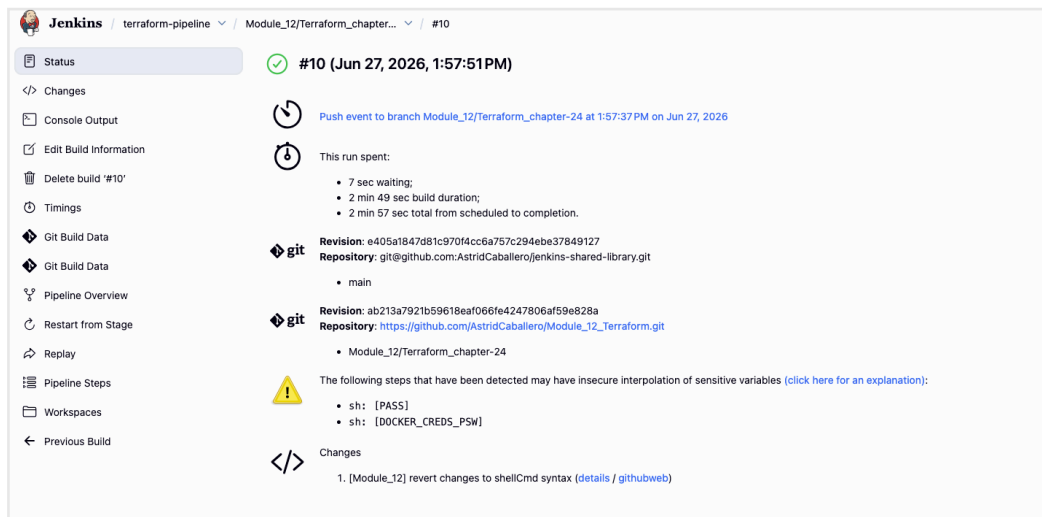


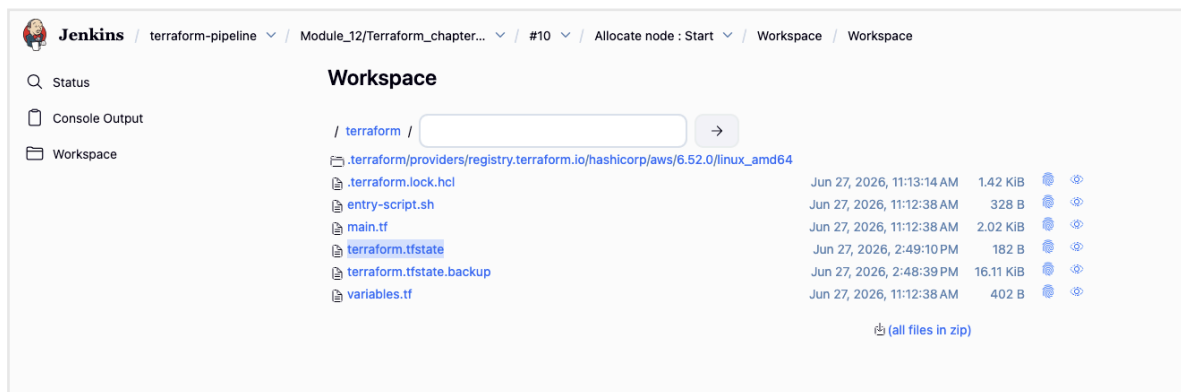
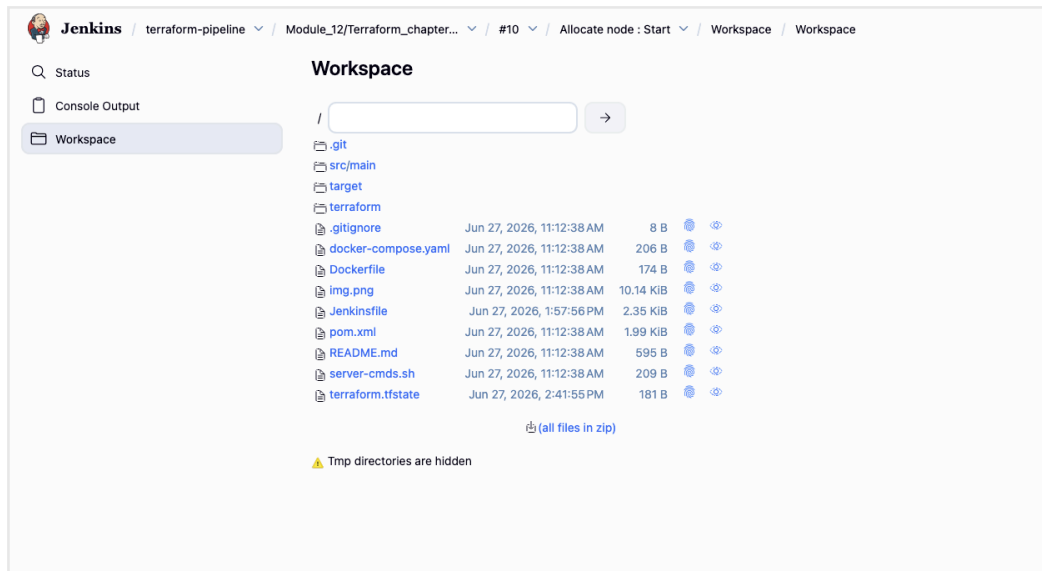
Another edition I made was:

Jenkinsfile sets the image name to version 2.0 so for consistency I updated the version 1.0 to 2.0 in the pom and dockerfile:



Also I destroyed the TF infra and for that I did it inside jenkins as is jenkins that holds the state of the infra called





1. So I ssh into my droplet

-> ssh root@165.22.116.21

for jenkins I am using the file id_ed25519 that is in .ssh file and this is the one by default so I don't need to specify it in the command above like I did to ssh into the EC2 from my laptop with 'myapp-key-pair.pem'

Q:

my laptop will use id_ed25519 by default right?

A:

Yes, exactly. When you run ssh user@host without -i, your SSH client tries your default keys in this order:

- ~/.ssh/id_ed25519
- ~/.ssh/id_rsa
- ~/.ssh/id_ecdsa

Since the DigitalOcean droplet was set up with your id_ed25519 public key, it just works automatically. No need to specify -i.

```

> ssh root@165.22.116.21
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jun 27 14:29:15 UTC 2026

System load:  0.24           Processes:           135
Usage of /:   16.6% of 76.45GB Users logged in:      0
Memory usage: 36%           IPv4 address for eth0: 165.22.116.21
Swap usage:   0%            IPv4 address for eth0: 10.16.0.6

Expanded Security Maintenance for Applications is not enabled.

51 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Last login: Fri Jun 26 09:43:00 2026 from 140.228.47.252
root@jenkins-server:~# ls

```

2. Then I got into the jenkins container as root user

-> docker ps

```

root@jenkins-server:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	NAMES	CREATED	STATUS	PORTS
56efcedae10e	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	naughty_dijkstra	2 weeks ago	Up 2 weeks	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp

-> docker exec -it -u 0 56efcedae10e bash

```

root@jenkins-server:~# docker exec -it -u 0 56efcedae10e bash
root@56efcedae10e:/# cd /var/jenkins_home/workspace/
root@56efcedae10e:/var/jenkins_home/workspace# ls
11_EKS_ECR_Full_CI_CD_chapter-10          java-maven-build
11_EKS_ECR_Full_CI_CD_chapter-10@tmp      java-maven-build@tmp
e_Module_12_Terraform_chapter-24          my-freestyle-job
e_Module_12_Terraform_chapter-24@libs     my-freestyle-job@tmp
e_Module_12_Terraform_chapter-24@tmp      workspaces.txt

```

3. Then I add the env vars

export AWS_ACCESS_KEY_ID=your-key

export AWS_SECRET_ACCESS_KEY=your-secret

```
_Terraform_chapter-24# export AWS_ACCESS_KEY_ID=AKIA2JG  
_Terraform_chapter-24# export AWS\_SECRET\_ACCESS\_KEY=Lscf
```

Q:

jenkins has the credentials stored via the UI why do I have to set them again as env vars?

A:

Good point. The AWS credentials stored in Jenkins UI are only injected into the environment when a pipeline job runs — Jenkins manages that injection through the credentials() binding in the Jenkinsfile.

When you manually exec into the container and run commands in a plain bash shell, you're outside of Jenkins' pipeline context so those credentials don't get injected automatically.

So you have two options:

Option 1 — Set env vars manually in the container (quick and dirty):

```
export AWS_ACCESS_KEY_ID=your-key  
export AWS_SECRET_ACCESS_KEY=your-secret  
terraform destroy -auto-approve
```

Option 2 — Trigger destroy through Jenkins (cleaner, credentials handled automatically):

- Add a manual destroy stage to your Jenkinsfile with an input step so it only runs when you approve it
- Or create a separate Jenkins job just for destroy

For a one-off cleanup Option 1 is fine, but Option 2 is the proper way to handle it in a real pipeline.

4. Then I cd into the workspace until I was inside the terraform folder

```

root@56efcedae10e:/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24# ls
Dockerfile          img.png             target
Jenkinsfile         pom.xml            terraform
README.md           server-cmds.sh     terraform.tfstate
docker-compose.yaml src                 terraform@tmp
root@56efcedae10e:/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24# cd ..
root@56efcedae10e:/var/jenkins_home/workspace# ls
11_EKS_ECR_Full_CI_CD_chapter-10      java-maven-build
11_EKS_ECR_Full_CI_CD_chapter-10@tmp   java-maven-build@tmp
e_Module_12_Terraform_chapter-24      my-freestyle-job
e_Module_12_Terraform_chapter-24@libs  my-freestyle-job@tmp
e_Module_12_Terraform_chapter-24@tmp   workspaces.txt
root@56efcedae10e:/var/jenkins_home/workspace# cd e_Module_12_Terraform_chapter-24
root@56efcedae10e:/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24# ls
Dockerfile          img.png             target
Jenkinsfile         pom.xml            terraform
README.md           server-cmds.sh     terraform.tfstate
docker-compose.yaml src                 terraform@tmp
root@56efcedae10e:/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24# cd terraform

```

5. Then I destroyed the infra
-> terraform destroy

```

root@56efcedae10e:/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24/terraform# terraform destroy -auto-approve
data.aws_ssm_parameter.latest-amazon-linux-image: Reading...
aws_vpc.myapp-vpc: Refreshing state... [id=vpc-0732451ce7b9bcb92]
data.aws_ssm_parameter.latest-amazon-linux-image: Read complete after 0s [id=/aws/service/ami-amazon-linux-latest/al2023-ami-l-default-x86_64]

```

And it is destroyed

```

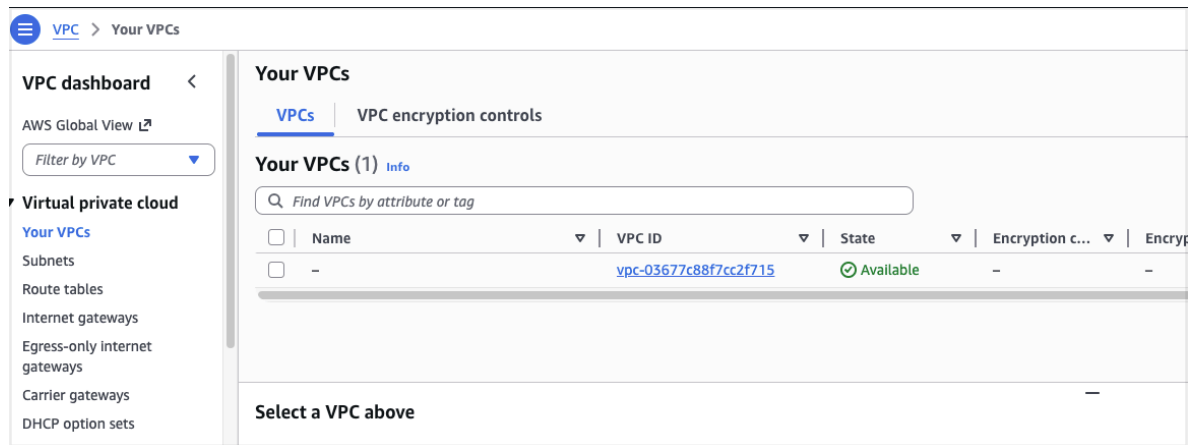
Plan: 0 to add, 0 to change, 6 to destroy.

Changes to Outputs:
  - ec2_public_ip = "18.171.52.103" -> null
aws_default_route_table.main-rtb: Destroying... [id=rtb-0f155ed517c5e7891]
aws_default_route_table.main-rtb: Destruction complete after 0s
aws_instance.myapp-server: Destroying... [id=i-039ea25c58f6c6e4a]
aws_internet_gateway.myapp-igw: Destroying... [id=igw-070cdb8340073620d]
aws_instance.myapp-server: Still destroying... [id=i-039ea25c58f6c6e4a, 00m10s elapsed]
aws_internet_gateway.myapp-igw: Still destroying... [id=igw-070cdb8340073620d, 00m10s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-039ea25c58f6c6e4a, 00m20s elapsed]
aws_internet_gateway.myapp-igw: Still destroying... [id=igw-070cdb8340073620d, 00m20s elapsed]
aws_internet_gateway.myapp-igw: Destruction complete after 27s
aws_instance.myapp-server: Destruction complete after 29s
aws_subnet.myapp-subnet-1: Destroying... [id=subnet-055f1bccf7ed23b2f]
aws_default_security_group.default-sg: Destroying... [id=sg-0a35ca1abb25221ce]
aws_default_security_group.default-sg: Destruction complete after 0s
aws_subnet.myapp-subnet-1: Destruction complete after 1s
aws_vpc.myapp-vpc: Destroying... [id=vpc-0732451ce7b9bcb92]
aws_vpc.myapp-vpc: Destruction complete after 1s

root@56efcedae10e:/var/jenkins_home/workspace/e_Module_12_Terraform_chapter-24/terraform#

```

In the AWS console now the VPC we created with jenkins is gone and I only have the default vpc



Questions

Q:

in jenkinsfile I have this code:

```

37     stage("provision server") {
38         environment {
39             AWS_ACCESS_KEY_ID = credentials('jenkins_aws_access_key_id')
40             AWS_SECRET_ACCESS_KEY = credentials('jenkins-aws_secret_access_key')
41             TF_VAR_env_prefix = 'test'
42         }
43         steps {
44             script {
45                 dir('terraform') {
46                     sh "terraform init"
47                     sh "terraform apply -auto-approve"
48                     EC2_PUBLIC_IP = sh(
49                         script: "terraform output ec2_public_ip",
50                         returnStdout: true
51                     ).trim()
52                 }
53             }
54         }
55     }

```

I wonder how Tf knows if the infra exists when I rebuild my pipeline so it doesn't create a new infra each time

A:

Terraform uses a state file (terraform.tfstate) to track what infrastructure already exists.

When you run terraform apply, Terraform:

1. Reads the state file to see what it already created
2. Compares it against your .tf code
3. Only creates/updates/destroys what has changed

So if nothing changed in your Terraform code, it will output something like:

No changes. Infrastructure is up-to-date.
And just return the existing `ec2_public_ip` output.

The important question is — where is your state file stored?

By default it's stored locally at `terraform/terraform.tfstate`. This causes a problem in Jenkins because:

- Each Jenkins build runs in a fresh workspace
- The state file from the previous build may not be there
- Terraform thinks nothing exists and creates duplicate infrastructure

The fix is a remote backend — store state in S3 so Jenkins always finds it:

```
hcl
terraform {
  backend "s3" {
    bucket = "your-bucket-name"
    key    = "terraform/state"
    region = "eu-west-2"
  }
}
```

Q:

it looks jenkins is not building new infra in my AWS as I have run the pipeline 3 times already with the provision of TF successful but deployment failure and I only have 1 vpc and 1 ec2 as expected

A:

That's because the `terraform.tfstate` file is being preserved in the Jenkins workspace between runs!

Jenkins reuses the same workspace directory (`/var/jenkins_home/workspace/your-pipeline-name`) for the same pipeline — it doesn't wipe it between builds. So the state file survives across runs and Terraform knows the infra already exists.

This is fine for now while you're learning, but it's worth knowing it's a bit fragile — if Jenkins is restarted, the workspace is cleaned, or you run on a different agent, the state file would be lost and you'd get duplicate infra. That's exactly why the S3 backend is the proper solution you'll learn next.

For now though, it's working in your favour!

Q:

how jenkins is configured to do this? where should I look?

A:

Jenkins does this by default — it's not something you configure, it's just how Jenkins works out of the box.

When a pipeline runs, Jenkins:

1. Looks at the pipeline name
2. Creates a workspace folder at `/var/jenkins_home/workspace/<pipeline-name>`
3. On the next run of the same pipeline, reuses that same folder

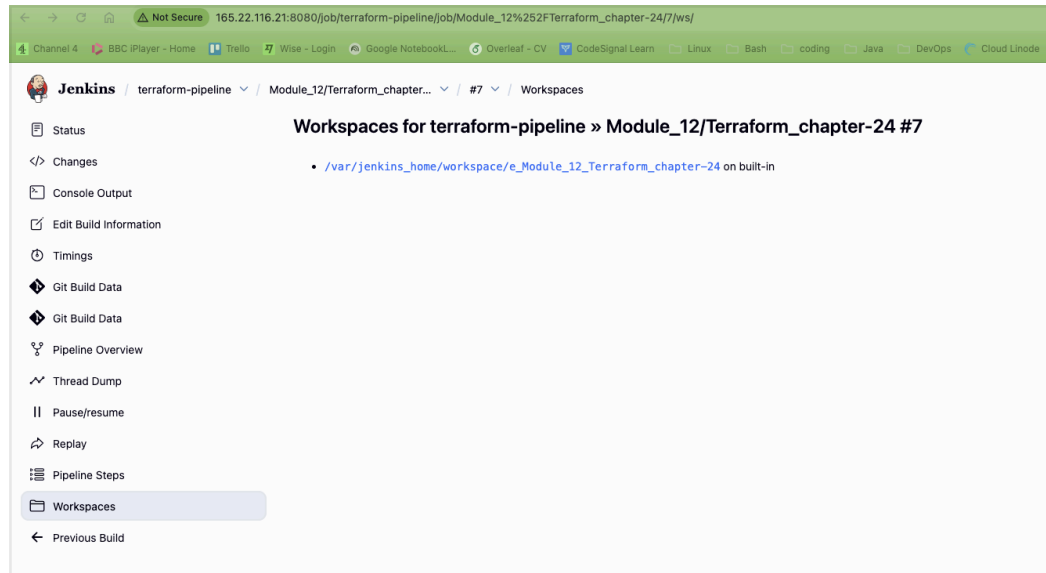
You can see this in action in your Jenkins UI:

- Open your pipeline → click on a build → click "Workspaces" in the left sidebar — it shows you the exact directory being used

Where you can change this behaviour:

- In the Jenkinsfile you can add `ws('custom-path')` to specify a different workspace
- In Jenkins UI → pipeline config → there's a option to wipe workspace before each build

But by default Jenkins intentionally reuses the workspace so things like your `terraform.tfstate` and downloaded dependencies persist between builds — it's a performance/convenience decision.



Q:

I will learn about s3 in the next lesson so for now how can I make sure to destroy the infra if the pipeline fails? so I don't have duplicate infra?

A:

You can add a post block in your Jenkinsfile that runs `terraform destroy` if the pipeline fails:

```

stage("provision server") {
    environment {
        AWS_ACCESS_KEY_ID = credentials('jenkins_aws_access_key_id')
        AWS_SECRET_ACCESS_KEY = credentials('jenkins-
aws_secret_access_key')
        TF_VAR_env_prefix = 'test'
    }
    steps {
        script {
            dir('terraform') {
                sh "terraform init"
                sh "terraform apply -auto-approve"
                EC2_PUBLIC_IP = sh(
                    script: "terraform output ec2_public_ip",
                    returnStdout: true
                ).trim()
            }
        }
    }
}

```

Then at the end of your full pipeline add:

```

post {
    failure {
        script {
            dir('terraform') {
                sh "terraform destroy -auto-approve"
            }
        }
    }
}

```

This way:

- Pipeline succeeds → infra stays up
- Pipeline fails → Terraform destroys everything it created

But also remember — before each run, manually check in AWS console that no leftover EC2/VPC resources exist from a previous failed run, just to be safe until you set up the S3 backend.